Approval gates and idempotency

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Planning and action selection. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of approval gates and idempotency. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **approval gate**s and idempotency.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **approval gate**s and idempotency

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Planning and action selection. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **approval gate**s and idempotency using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Approval gates and idempotency case. Separate observations, given facts and assumptions before choosing an explanation.

Approval gates bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Approval gates and idempotency. Connect each labelled input, deciding step and result.

The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

Approval gates and idempotency — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload.	The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.	Return the stored committed result; one effect total. A timeout alone is not evidence the first effect failed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Approval gates and idempotency. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: A new K10 asks for desk 5 without approval.

> **Predicted result**
>
> **Prediction**: Stop K10 at the **approval gate**.

> **Why the result follows**
>
> Reason: K9's approval does not apply to a different action.

Words that unlock the idea

> Use the vocabulary of Approval gates and idempotency to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Idempotency key	An identifier for deduplicating one logical operation
Approval gate	A required permission check before a side effect
Payload conflict	Different operation content under the same key

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Approval gates and idempotency, predict a step, then compare with its explanation.

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload.

Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Worked explanation · Approval gates and idempotency

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Approval gates and idempotency

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from copy import deepcopy

class MockLedger:
    """In-memory classroom deduplication, not durable exactly-once delivery."""

    def __init__(self):
        self.committed = {}
        self.effects = []

    def submit(self, key, payload, approval):
        if approval != payload:
            raise PermissionError('exact approval required')
        if key in self.committed:
            (original, result) = self.committed[key]
            if original != payload:
                raise ValueError('idempotency conflict')
            return deepcopy(result)
        result = {'operation_id': key, 'status': 'committed'}
        self.effects.append(deepcopy(payload))
        self.committed[key] = (deepcopy(payload), deepcopy(result))
        return deepcopy(result)
```

```
ledger = MockLedger()
payload = {"version": "v1", "destination": "D"}
ledger.submit("K", payload, payload)
ledger.submit("K", payload, payload)
print(len(ledger.effects))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Approval gates and idempotency

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

The facts we will use

> **Supplied facts**
>
> K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

AI ACADEMY / AI-6/15

Approval gates and idempotency

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check approval

The approved identifier and desk match K9.



2. Execute first copy

No result is stored yet, so perform the approved reservation once.



3. Handle duplicate

Return the stored K9 result without another reservation.

CHANGE ONE THING

A new K10 asks for desk 5 without approval.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check approval	The approved identifier and desk match K9.
2. Execute first copy	No result is stored yet, so perform the approved reservation once.
3. Handle duplicate	Return the stored K9 result without another reservation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

What this example does not establish

Keep the stated limit

Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

Keep the stated limit

Read the labelled picture: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?

Keep the stated limit

Change one thing: A new K10 asks for desk 5 without approval. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Approval gates and idempotency claim and explain a repair using the supplied evidence.

A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

Check the claim against the facts

A report omits this qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Approval gates and idempotency with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Approval gate**s and idempotency". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Approval gate**s and idempotency. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Approval gates and idempotency. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

```
from copy import deepcopy

class MockLedger:
    """In-memory classroom deduplication, not durable exactly-once delivery."""

    def __init__(self):
        self.committed = {}
        self.effects = []
```

```
    def submit(self, key, payload, approval):
        if approval != payload:
            raise PermissionError('exact approval required')
        if key in self.committed:
            (original, result) = self.committed[key]
            if original != payload:
                raise ValueError('idempotency conflict')
            return deepcopy(result)
        result = {'operation_id': key, 'status': 'committed'}
        self.effects.append(deepcopy(payload))
        self.committed[key] = (deepcopy(payload), deepcopy(result))
        return deepcopy(result)

ledger = MockLedger()
payload = {"version": "v1", "destination": "D"}
ledger.submit("K", payload, payload)
ledger.submit("K", payload, payload)
print(len(ledger.effects))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Approval gates and idempotency. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the

> evidence supporting it.

Add a labelled **approval gate**s and idempotency **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **approval gate**s and idempotency.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Approval gates and idempotency case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Approval gates and idempotency — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload.	The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.	Return the stored committed result; one effect total. A timeout alone is not evidence the first effect failed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

> Key J already committed room A. A retry uses key J but room B. What should happen?

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

> Key J already committed room A. A retry uses key J but room B. What should happen?

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Approval gates and idempotency, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

By the end, I can	How I will show it
Explain the mechanism and terminology of approval gates and idempotency	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Approval gates and idempotency without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **approval gate**s and idempotency

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the stored K9 result without another reservation.

1. Execute first copy

2. Check approval

3. Handle duplicate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Handle duplicate Return the stored K9 result without another reservation. Approval answers whether an action may proceed.

Idempotency prevents repeated handling of that action from creating duplicate effects.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A new K10 asks for desk 5 without approval. Which conclusion is supported by the supplied facts?

1. Stop K10 at the **approval gate**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop K10 at the **approval gate**.

> K9's approval does not apply to a different action.

Repair a missing building block

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

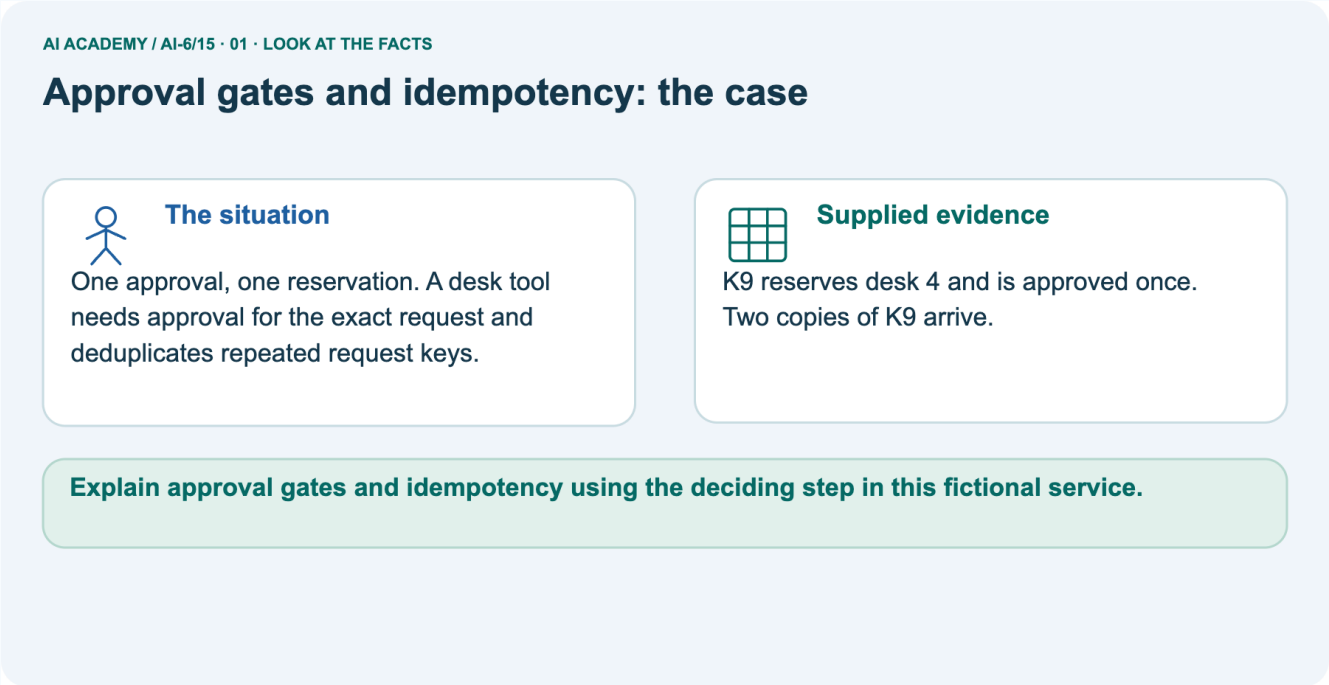
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Approval gates and idempotency graphic using its labels, deciding evidence and relevant limitation.

01. Approval gates and idempotency: the case

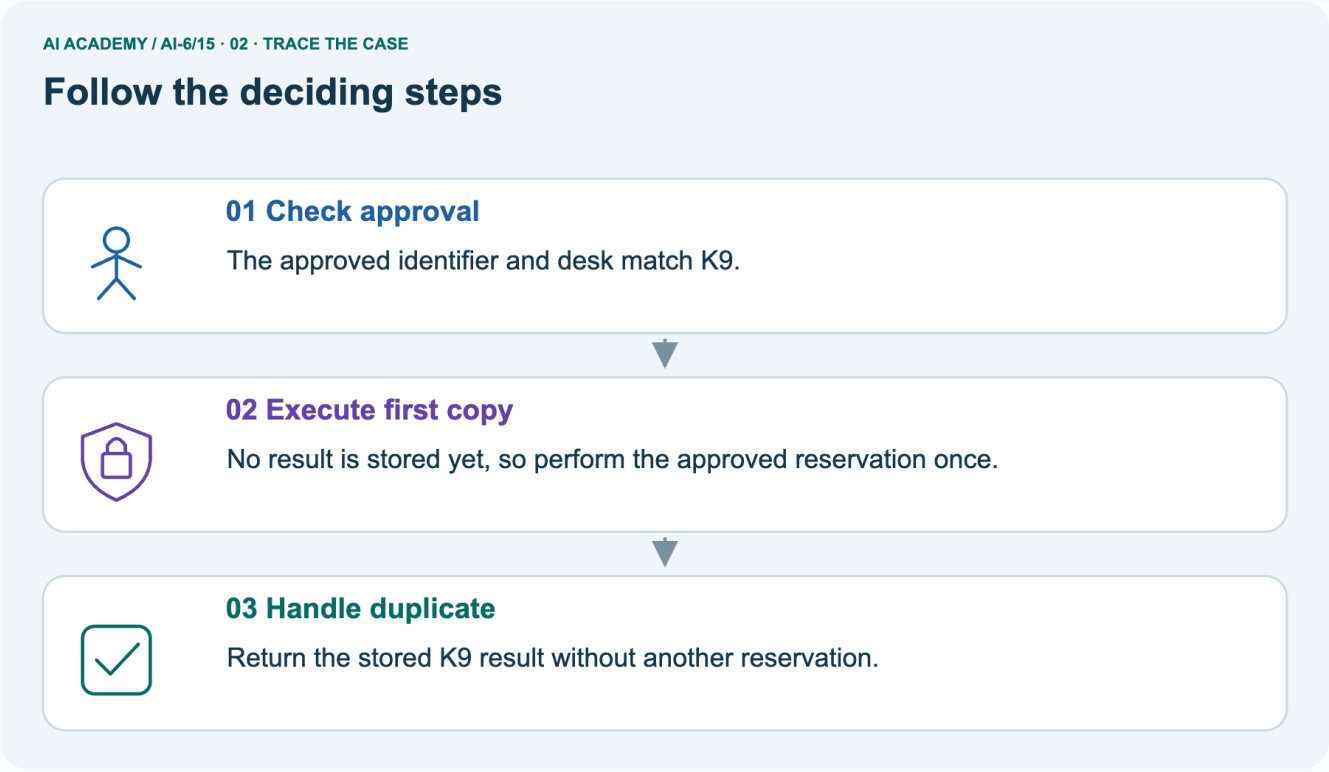


Follow the labelled picture with the matching explanation. Open the vector diagram

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys. Supplied facts: K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

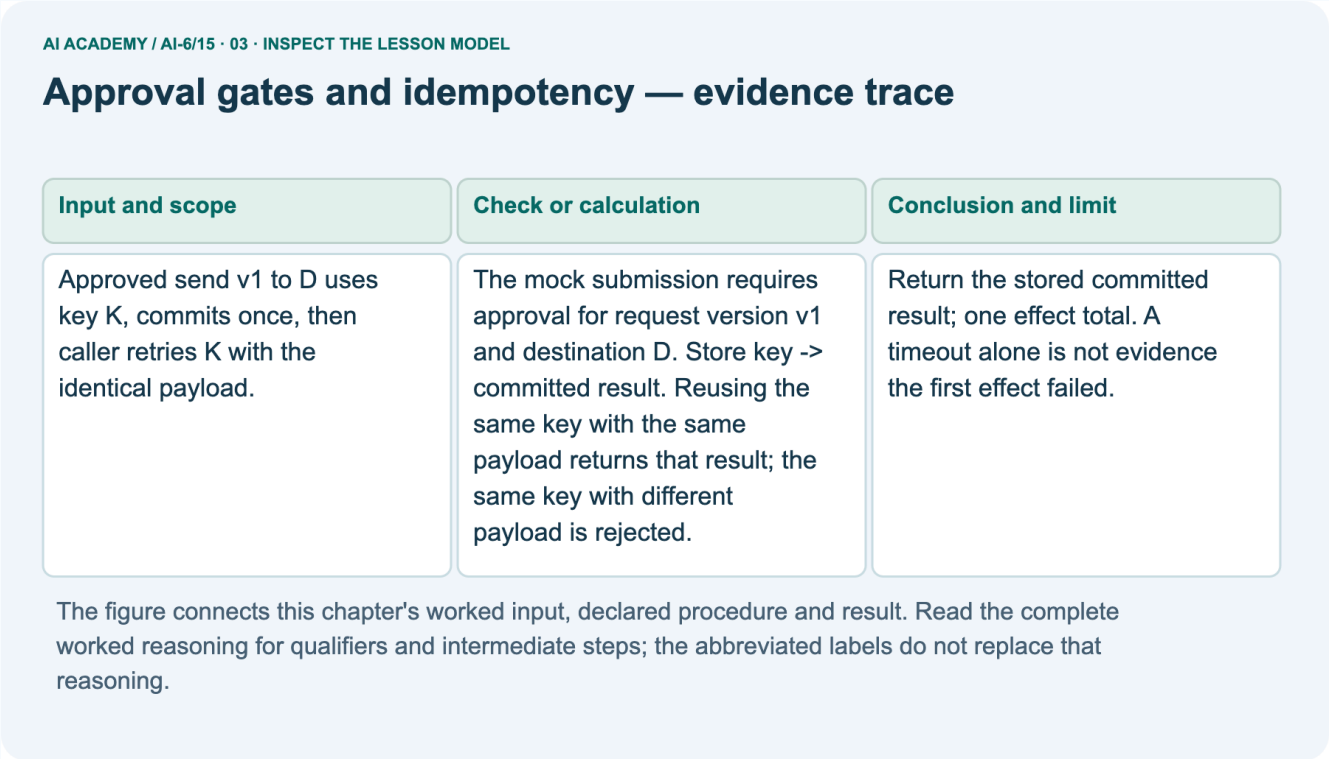


Follow the labelled picture with the matching explanation. Open the vector diagram

Check approval: The approved identifier and desk match K9. → Execute first copy: No result is stored yet, so perform the approved reservation once. → Handle duplicate: Return the stored K9 result without another reservation.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Approval gates and idempotency — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

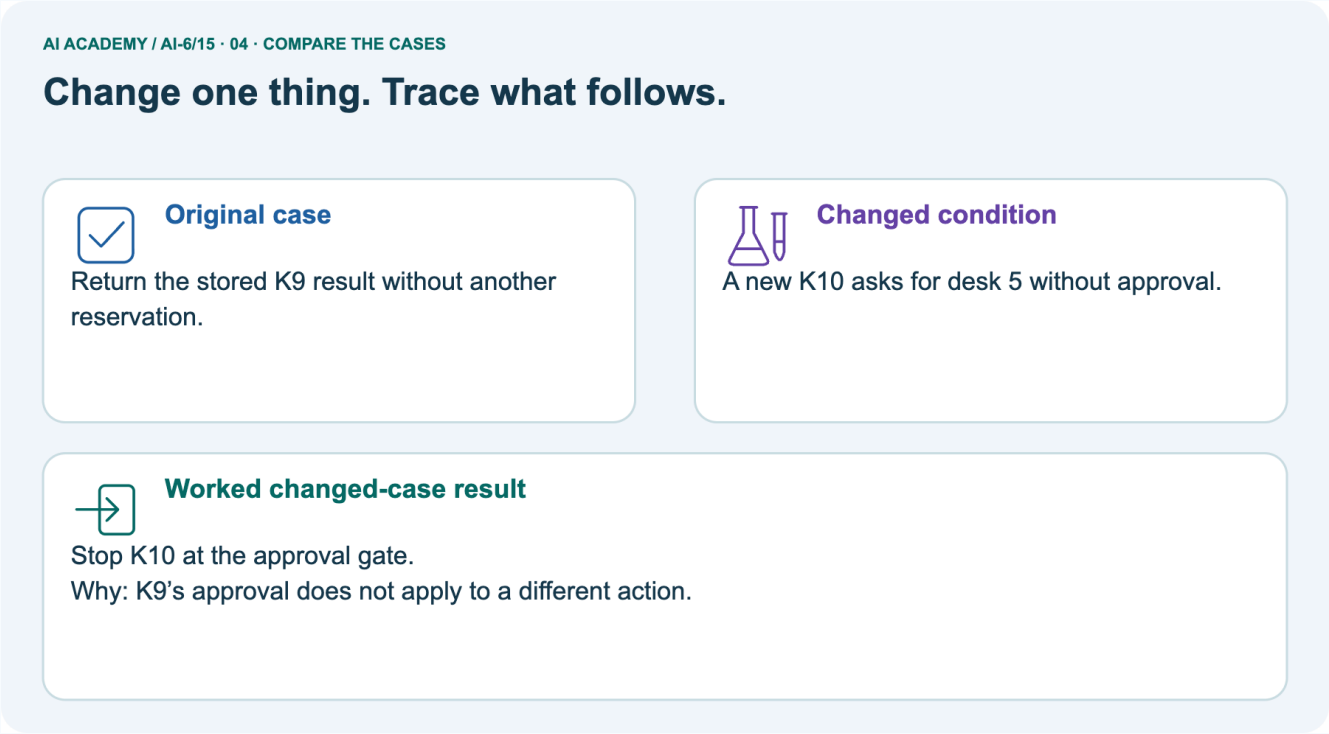
Approval gates and idempotency — **evidence** trace.

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. | The mock submission requires approval for request version v1 and destination D. Store key -> committed result.

Reusing the same key with the same payload returns that result; the same key with different payload is rejected. | Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



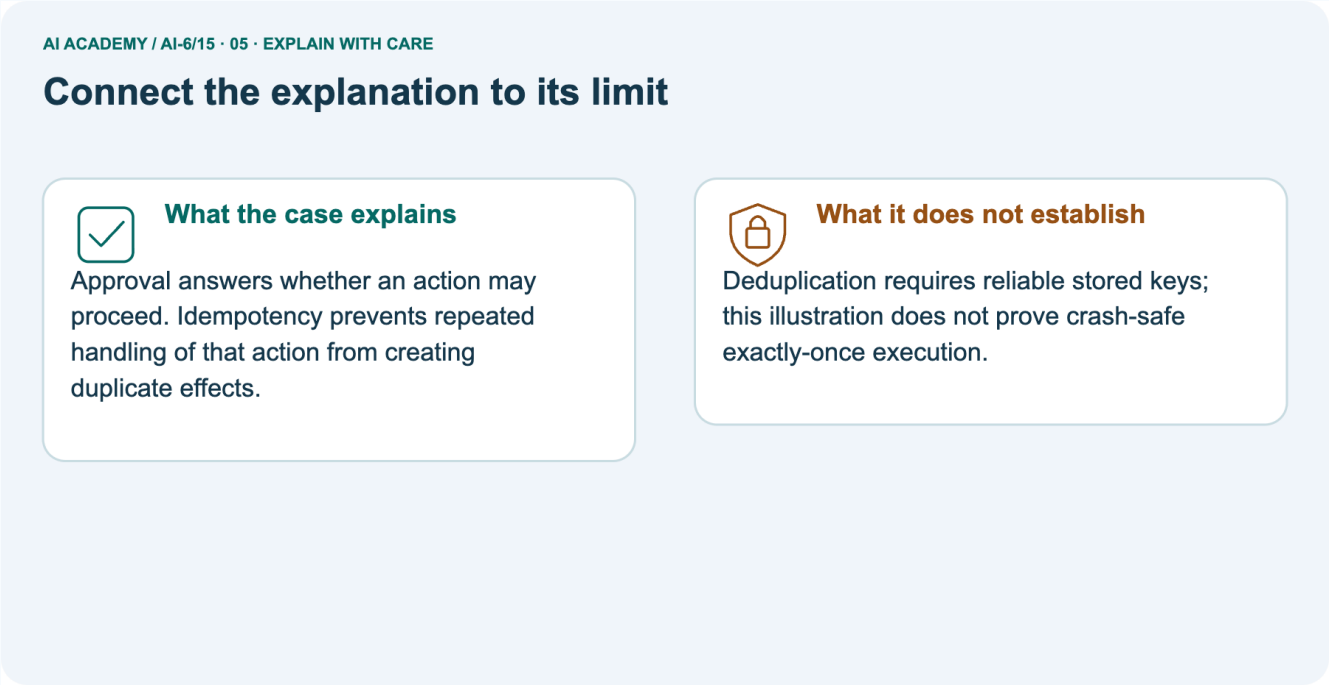
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return the stored K9 result without another reservation. Changed condition: A new K10 asks for desk 5 without approval. Expected result: Stop K10 at the **approval gate**.

Reason: K9's approval does not apply to a different action.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



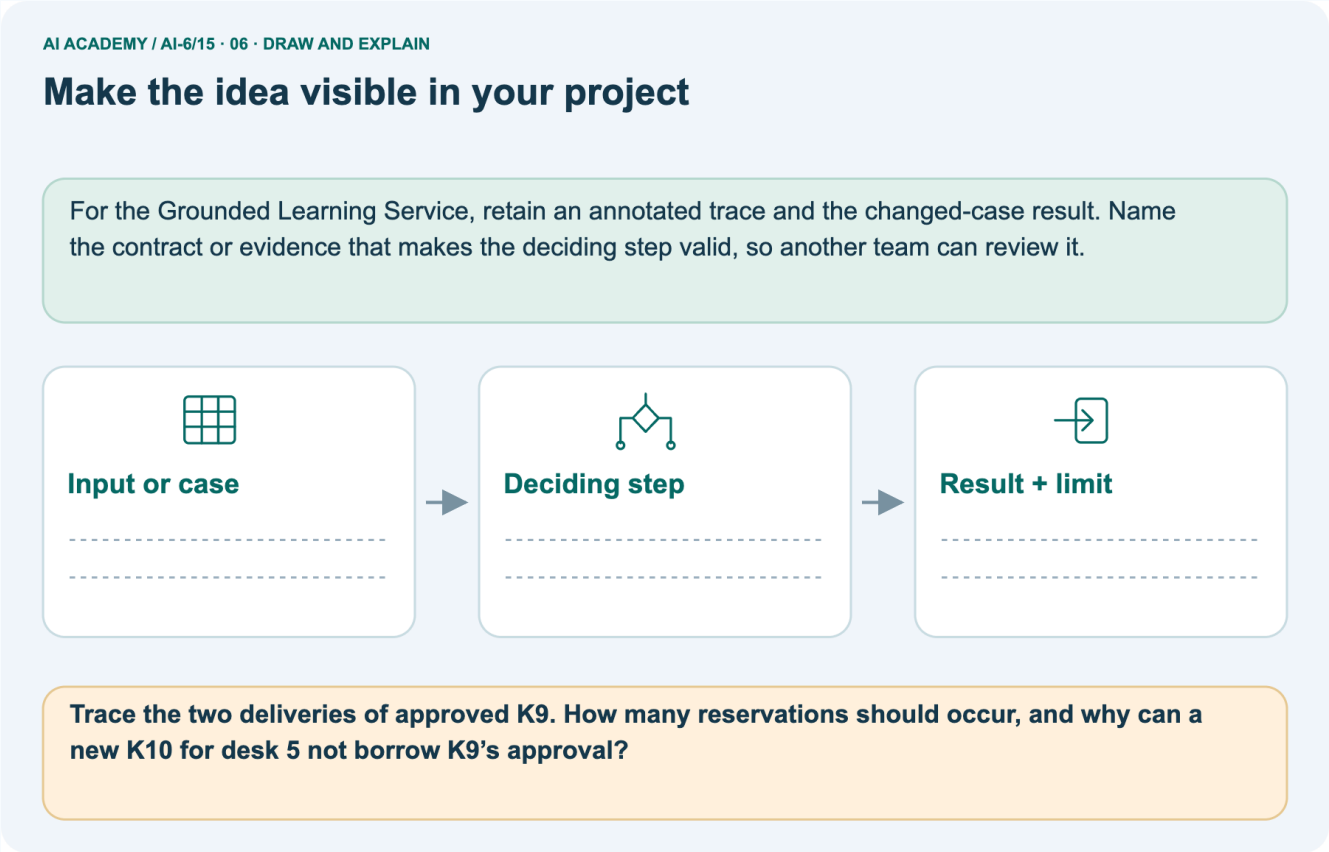
Follow the labelled picture with the matching explanation. Open the vector diagram

Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

Limit: Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.